

TextMonkey: An OCR-Free Large Multimodal Model for Understanding Document

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Abstract—We present TextMonkey, a large multimodal model (LMM) tailored for text-centric tasks. Our approach introduces enhancement across several dimensions: By adopting Shifted Window Attention layer, we achieve cross-window connectivity at higher input resolutions and stabilize early training; We hypothesize that images may contain redundant tokens, and by using similarity to filter out significant tokens, we can not only streamline the token length but also enhance the model’s performance. Moreover, by expanding our model’s capabilities to encompass text spotting and grounding, and incorporating positional information into responses, we enhance interpretability. Evaluation on 12 benchmarks shows notable improvements: 5.2% in Scene Text-Centric tasks (including STVQA, TextVQA, and OCRVQA), 6.9% in Document-Oriented tasks (such as DocVQA, InfoVQA, ChartVQA, DeepForm, Kleister Charity, and WikiTableQuestions), and 2.8% in Key Information Extraction tasks (comprising FUNSD, SROIE, and POIE). It outperforms in scene text spotting with a 10.9% increase and sets a new standard on OCRBench, a comprehensive benchmark consisting of 29 OCR-related assessments, with a score of 561, surpassing previous open-sourced large multimodal models for document understanding.

Index Terms—Large multi-modal model, document analysis, scene text, resolution, OCRBench.

I. INTRODUCTION

EXTRACTING key information from a variety of sources, including documents like tables, forms, and invoices, as well as text in the wild is crucial for industries and academic research, aiming to automate and refine document-based and scene-text workflows. This field requires text detection and recognition in both document images and real-world scenes, language comprehension, and the integration of vision and language.

Many early methods [1], [2] attempt to address the task using a two-stage approach: 1) Detect and recognize the text using external systems; 2) Document understanding based on the fusion of text results and images. However, the individual

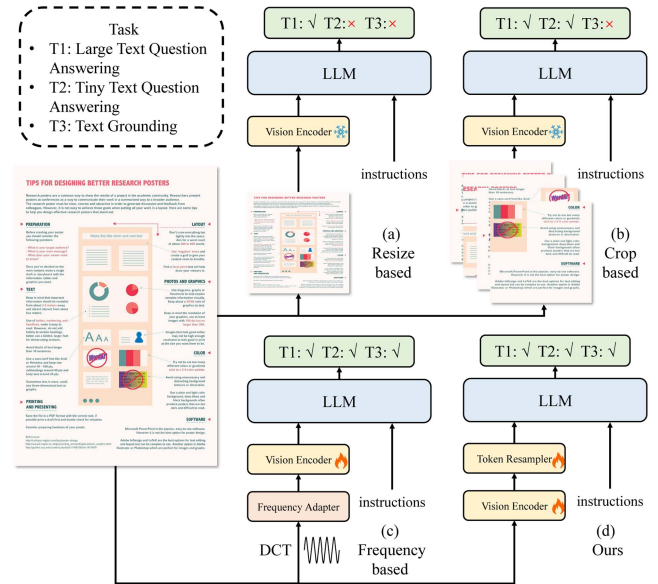


Fig. 1. Comparisons to the existing pipelines for document understanding. Compared to (a) Resize based methods, (b) Crop based methods, and (c) frequency based methods, our model can efficiently process high-resolution text-related images with various tasks.

step of text reading in the processing pipeline may lead to the accumulation of errors. Moreover, relying on off-the-shelf OCR Models/APIs (OCR-Models) introduces additional engineering complexity, limits the connection between the text and its surrounding context, and can potentially increase computational costs. To alleviate the drawbacks of external systems before understanding, OCR-Free solutions [3], [4] have attracted increasing attention recently.

The field of large multimodal models (LMMs) [5], [6] is advancing rapidly due to its powerful ability to handle diverse types of data. However, they still have limitations when it comes to addressing text-related tasks. As depicted in Fig. 1(a), several methods, including LLaVAR [7] and UniDoc [8] heavily rely on a pre-trained CLIP [9] for visual encoding. Nevertheless, these encoders have input resolutions of 224 or 336, which are insufficient to meet the demands of documents containing numerous small texts [10]. Therefore, they can only recognize large text and struggle with small text in images. To address the limitations of tiny text, UReaer [11] and Monkey [12] take a cropping strategy to expand the input resolution, as shown in Fig. 1(b). However, this crop strategy may inadvertently split

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Code is released at <https://github.com/Yuliang-Liu/Monkey>.

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related words, resulting in semantic incoherence. For example, the word "Backup" may be divided into "Back" and "up," making it impossible to restore its original meaning even after fusion has been performed. Besides, the spatial separation caused by this splitting also makes it challenging to handle text position-related tasks, such as text grounding. As shown in Fig. 1(c), DocPedia [13] directly processes visual input in the frequency domain rather than the pixel space. Due to the characteristics of the frequency domain, it can quickly expand the resolution without losing information. However, due to the transformation of the feature space, it is difficult to leverage existing pretrained models, increasing the demand for training resources.

We want to inherit the efficient image resolution scaling feature of Monkey [12] but address the missing cross-window context for the documents mentioned above. For this purpose, we introduce TextMonkey, as shown in Fig. 1(d). TextMonkey utilizes a Split Module that divides high-resolution images into window patches using a sliding window method. Inspired by [14], we treat every self-attention layer in the CLIP as self-attention in non-overlapped windows. To introduce cross-window relationships while maintaining efficient computation, we use Shifted Window Attention with zero-initialization to build cross-window connections. This approach allows us to maintain the training data distribution for the encoder and deal with high-resolution document images while reducing the computational cost of training from scratch. On the other hand, the utilization of the Split Module still poses a significant challenge as it leads to a notable increase in token length. We have observed that there are numerous repetitive image features that align with the language space, similar to certain repeated elements in the language itself. Thus, we propose a token resampler to compress these features while keeping as many of the most important features as possible. We employ important tokens as queries and the original features as key-value pairs, facilitating the reaggregation of features. On the basis of reducing the number of tokens, our module can also significantly improve the performance compared to random queries.

On the other hand, due to the self-explanatory nature of the text, in most cases, humans are able to locate the position of the answer itself. To alleviate the issue of hallucination in large language models further, we require the model not only to provide accurate answers but also to locate specific visual evidence supporting its response. We also introduce a variety of text-related tasks to deepen the connection between text information and visual information, such as text spotting and text grounding. Besides, incorporating positional cues into the answers can further enhance the model's reliability and interpretability.

We summarize the advantages of our method as follows:

- *Enhancing cross-window relations:* We adopt Shifted Window Attention to successfully incorporate cross-window connectivity while expanding the input resolutions. Besides, we introduce zero initialization in the Shifted Window Attention mechanism, enabling the model to avoid drastic modifications to early training.

- *Token compression:* We show enlarging resolution results in some redundant tokens. By using similarity as a criterion, we are able to find significant tokens that serve as queries for the token resampler. This module not only reduces the token length but also improves the model's performance. Additionally, it significantly improves the performance compared to the use of random queries.
- *Support text grounding:* We expand our scope to include tasks beyond text-based question answering, encompassing reading text, text spotting, and text grounding. Additionally, we found that incorporating positional information into the answers can improve the model's interpretability.
- We evaluated TextMonkey's performance across 12 recognized benchmarks, observing significant improvements in several areas. Firstly, in scene text-centric tasks such as STVQA, TextVQA, and OCRVQA, TextMonkey achieved a 5.2% increase in performance. For document-oriented tasks, including DocVQA, InfoVQA, ChartVQA, DeepForm, Kleister Charity, and WikiTableQuestions, it showed a 6.9% improvement. In the domain of key information extraction tasks, like FUNSD, SROIE, and POIE, we noted a 2.8% uplift. Particularly notable was its performance in scene text spotting tasks (Total-Text, CTW1500, and ICDAR 2015) focused on transcription accuracy, where it improved by 10.9%. Additionally, TextMonkey set a new high score of 561 on OCRBench, a comprehensive benchmark encompassing 29 OCR-related evaluations, significantly surpassing the performance of previous open-source, large-scale multimodal models designed for document understanding. This achievement underscores TextMonkey's effectiveness and advances in the field of document analysis and understanding.

II. RELATED WORKS

Models designed to comprehend images with text information can be broadly categorized into two types: OCR-Model-Driven and OCR-Free methods.

A. OCR-Model-Driven Methods

OCR-Model-Driven methods use OCR tools to acquire text and bounding box information. Subsequently, they rely on the models to integrate text, layout, and visual data. Meanwhile, diverse pre-training tasks are devised to enhance cross-modal alignment between visual and text inputs. StrucTexT [15] pays attention to the fine-grained semantic information and global layout information within the image in the design of pre-training tasks. Based on layout knowledge enhancement technology, ERNIE-Layout [16] innovatively proposes two self-supervised pre-training tasks: reading order prediction and fine-grained image-text matching. The LayoutLM [2], [17], [18] series continuously improves by integrating pre-trained text, layout, and visual features and introducing a unified model architecture and pre-training goals. This enhances the model's performance in various document understanding tasks and simplifies overall

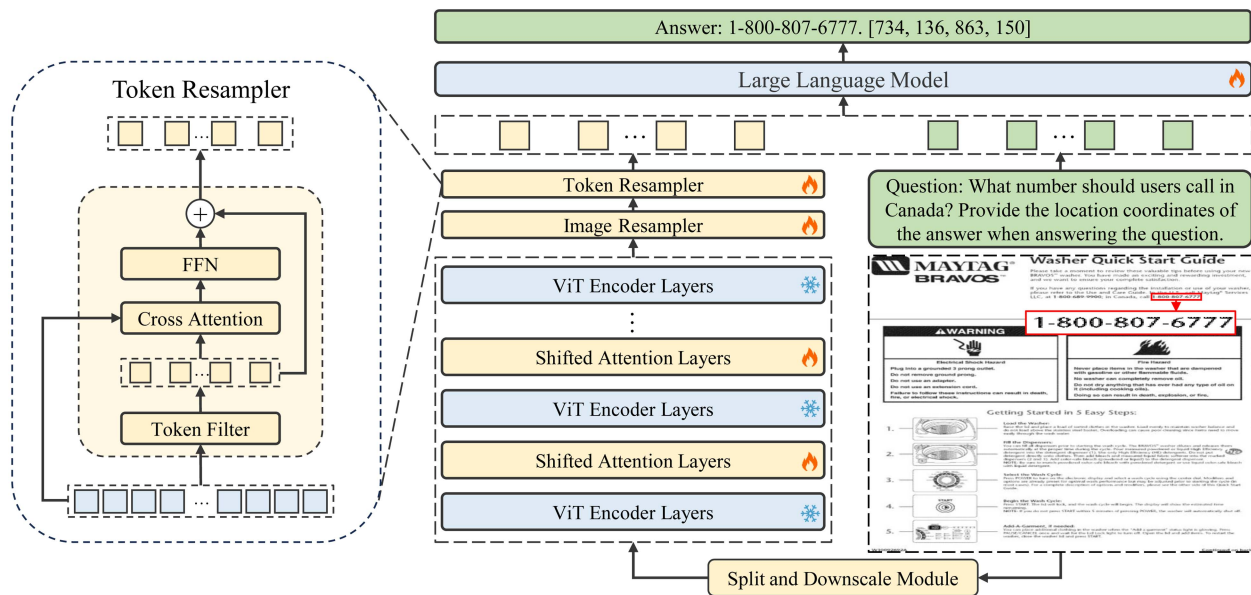


Fig. 2. An overview of the TextMonkey. It enables the enhancement of resolution with limited training resources while preserving cross-window information and reducing redundant tokens introduced by resolution enhancement.

design. UDOP [1] unifies vision, text, and layout through VTL Transformer and a unified generative pre-training task. Wukong-reader [19] proposes the Textline-Region Contrastive Learning and specially crafted pre-training tasks to extract fine-grained text line information. DocFormerv2 [20] designs an asymmetric pre-training method and a simplified visual branch for visual document understanding.

While advancements have been achieved, OCR-Model-Driven methods depend on text extraction from external systems, which necessitates increased computational resources and extends processing durations. Additionally, these models may inherit OCR inaccuracies, presenting challenges to document understanding and analysis tasks.

B. OCR-Free Methods

Donut [3] is the first to propose an OCR-free, Transformer-based model, while Dessurt [21] further introduces a two-way cross-attention mechanism to enhance the architecture. Pix2Struct [4] and StrucTexTv2 [22] propose corresponding pre-training schemes tailored for webpage screenshots and document images, respectively. Although these methods are free from the limitations of OCR engines/APIs, they still require fine-tuning for specific tasks. With the rapid advancement of multi-modal large language models, LLaVAR [7] and UniDoc [8] construct instruction-tuning datasets tailored for text-centric images to enhance text-related tasks. UReader [11] introduces a shape-adaptive cropping module to improve performance on high-resolution document images. DocPedia [13] supports higher resolutions by processing visual inputs in the frequency domain. Vary [23] extends input resolution by training a visual vocabulary on large-scale data, while TGDdoc [24] further incorporates text-grounding tasks. Qwen2.5-VL [25] and Qwen3-VL [26] pretrain vision encoders that natively support high-resolution

inputs without image tiling, but at the cost of substantial training resources. More recently, OCR-oriented MLLMs such as dots.ocr [27], MinerU 2.5 [28], and DeepSeekOCR [29] are specifically optimized for document parsing and text extraction; however, they are largely limited to standardized inputs, such as scanned or well-formatted PDF files. In contrast, TextMonkey supports a broader range of document and scene text understanding tasks, and is capable of handling photographed document images as well as natural scene text images.

III. METHODOLOGY

The method presented in Fig. 2 begins by dividing the input image into non-overlapping patches using a sliding window module, with each patch sized at 448×448 pixels. These patches are further subdivided into smaller patches of 14×14 pixels, each considered as a token. Utilizing Transformer blocks that inherit from the pre-trained CLIP model, we process these tokens on each window patch separately. To establish connections among various window patches, Shifted Window Attention is integrated at specific intervals within the Transformer blocks. To generate a hierarchical representation, the input image is resized to 448×448 and fed into CLIP to extract a global feature, as suggested by [12]. This global feature, alongside features from sub-images, is then processed by a shared image resampler to align with the language domain. Then, a Token Resampler is employed to further minimize redundancy in the language space by compressing the length of tokens. Ultimately, these processed features, combined with the input question, are analyzed by a Large Language Model (LLM) to produce the required answers.

A. Shifted Window Attention

Previous studies have underscored the significance of input resolution for precise document understanding [10], [13]. To

enhance training efficiency, recent methods [11], [12] have adopted a sliding window technique to enhance image resolution. While effective in analyzing natural scenes due to their localized content, this strategy may lead to the fragmentation of connected text in document analysis, disrupting semantic continuity. Additionally, the spatial disjunction poses challenges for tasks that rely on text positioning, such as text grounding.

To alleviate the issue mentioned above, we adopt Shifted Window Attention [14] to augment the CLIP model's visual processing capabilities. Unlike the original Swin Transformer, which primarily reduces the computational and memory cost of global self-attention via window partitioning and shifting, our Shifted Attention layers are designed to address the lack of semantic connections across image patches in document images. By applying Shifted Attention to the ViT features of different patches, we effectively capture contextual dependencies among all patches, thereby enhancing global semantic consistency. Specifically, for an input image $I \in \mathbb{R}^{H \times W \times 3}$, our approach slices the image into non-overlapping windows. This slice is achieved using a sliding window $W \in \mathbb{R}^{H_v \times W_v}$, where H_v and W_v indicate the window's size. Within each window, we independently apply a transformer block from the CLIP architecture, which initially does not account for cross-window relationships. To incorporate interactions between different windows and enhance the model's contextual understanding of the image, we adopt the Shifted Window Attention (SWA) mechanism. As mentioned in [14], the sliding window is cyclic-shifting toward the top-left direction, resulting in new windows. The self-attention is computed using a masking mechanism, which restricts the computation to tokens within each shifted window.

To achieve smoother training initialization, we have made modifications to the shifted window attention by allowing them to start learning from zero initialization, avoiding excessive transformation of early features during the initial stages. In particular, we modify the regular initialization in MLP to zero initialization to achieve smoother training, inspired by [30]:

$$x = \mathbf{B}\mathbf{A}\hat{x}, \quad (1)$$

where $\mathbf{B}$ and $\mathbf{A}$ refer to the weight of two linear layers. We use a random Gaussian initialization for $\mathbf{A}$ and zero initialization for $\mathbf{B}$. This approach ensures that the image encoder's parameters remain stable in the initial phase, facilitating a smoother training experience.

B. Image Resampler

To reduce the redundancy in image features initially, we inherited the image resampler from Qwen-VL [31], which is applied to every window. The module employs a set of trainable parameters as query vectors and utilizes the image features from the visual encoder as keys and values for cross-attention operations. This process helps compress the visual feature sequence to a fixed length of 256. Furthermore, to preserve positional information crucial for fine-grained image comprehension, 2D absolute positional encodings are integrated into the query-key pairs of the cross-attention mechanism.

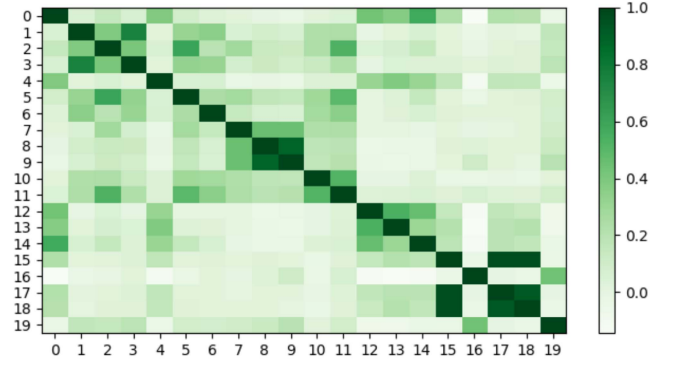


Fig. 3. Image token similarity comparisons. We randomly select 20 ordered tokens from image tokens and use cosine similarity as the metric for measuring similarity.

C. Token Resampler

As the image resolution increases, the crop-based method produces more image patches, resulting in a larger number of visual tokens. However, due to the limited input length of language models and the constraints of training cost, it becomes essential to reduce the number of tokens while preserving key information. Previous work [32] has shown the feasibility of merging redundant tokens in visual tasks. Similarly, when the resolution of an image increases, redundant visual information often appears. To quantify this redundancy, we compute the similarity between the image tokens. The similarity between two tokens t_i and t_j is defined as:

$$\text{Sim}(t_i, t_j) = \frac{t_i \cdot t_j}{\|t_i\| \|t_j\|}. \quad (2)$$

We randomly select 20 ordered tokens after the image resampler and compute pairwise similarities, as shown in Fig. 3. The visualization reveals that many tokens are highly similar, indicating the presence of substantial redundancy at higher resolutions. We further analyze the redundancy ratio at different resolutions by counting the number of tokens with similarity greater than a threshold $\tau = 0.8$. As shown in Fig. 4, the proportion of redundant tokens increases from 26.6% at 448^2 resolution to 55.8% at 896^2 , and 59.5% at 1334^2 . This confirms that increasing resolution not only enhances clarity but also introduces repetitive visual patterns.

To effectively compress redundant tokens, we introduce a Token Resampler that identifies and preserves the most informative tokens. Our key observation is that some tokens are highly unique and lack closely similar counterparts (e.g., the fifth token in Fig. 3), suggesting that they encode distinctive visual semantics. To formalize this intuition, we define an importance score for each token based on its maximum similarity with others:

$$s_i = \max_{j \neq i} \text{Sim}(t_i, t_j), \quad \text{Imp}(t_i) = 1 - s_i. \quad (3)$$

We hypothesize that the most representative tokens are those exhibiting minimal redundancy. Accordingly, tokens with higher importance scores $\text{Imp}(t_i)$ are more distinctive and tend to encode key information. We then retain the top- r tokens according

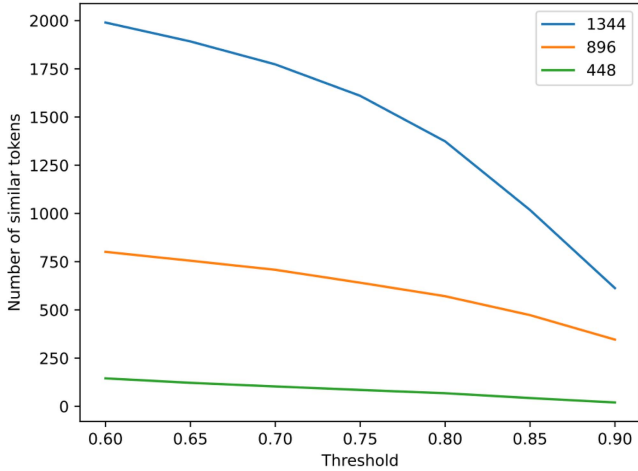


Fig. 4. Quantitative analysis of redundant tokens. The x-axis denotes the similarity threshold, while the y-axis shows the number of redundant tokens under different resolutions.

to their importance scores:

$$T_r = \text{Top-}r(\{t_i\}, \text{Imp}(t_i)). \quad (4)$$

The selected tokens are reordered according to their original spatial order to maintain positional consistency. To avoid potential information loss caused by discarding the remaining tokens, we further employ a cross-attention mechanism that allows the selected tokens to aggregate information from all tokens. To avoid potential information loss caused by discarding tokens, we apply a cross-attention mechanism. Let all image tokens be (T) and the top- r selected tokens be (T_r). The cross-attention is then formulated as:

$$T_r = \text{CrossAttn}(Q = T_r, K = T, V = T). \quad (5)$$

This operation ensures that each selected token attends to all tokens, effectively capturing global context while maintaining a compact token representation. Through this adaptive resampling mechanism, our module significantly reduces token redundancy and improves efficiency. Unlike Flamingo’s [33] Resampler, our Token Resampler employs a similarity-based filtering mechanism to retain the most representative tokens and suppress redundant ones. The model then aggregates information from all tokens via cross-attention, producing a more efficient image representation. In contrast, Flamingo relies on learnable queries that compress token representations to a fixed length.

D. Position-Related Task

To alleviate the issue of hallucinations in Large Language Models (LLMs), where they can produce incorrect responses not related to the provided image, we aim to enhance their capability to analyze and incorporate visual information into their replies. Considering that answers to text-based tasks are often found within the image itself, we anticipate that the large model will not only produce precise responses but also identify the particular visual proof that underpins its answer.

Moreover, we have undertaken modifications to existing question-answering datasets. Specifically, we have found the

TABLE I
PROMPTS FOR A VARIETY OF TASKS

Type	Prompt
Read All Text	Read all the text in the image.
Text Spotting	OCR with grounding:
Original Tasks	{Question}. Answer:
Position of text	<ref>text</ref>
Text Recognition	<ref>This</ref> <box>(x1,y1),(x2,y2)</box>is
VQA Grounding	{Question}. Provide the location coordinates of the answer when answering the question.

positions with the majority of answers in the images. These positional cues have been extracted and seamlessly integrated into the answers themselves. To preserve the original capability of direct dialogue, we have also retained the original question-answering task.

For better perception of the spatial positions of the text, it requires the model to have a strong spatial understanding. Building upon the aforementioned model designs, we add additional training tasks to improve the model’s perception of text positions, such as text spotting and reading text. Specific tasks and prompts are shown in Table I. Specifically, Read All Text requires the model to read all text in the image; Text Spotting detects text locations and recognizes their content; Original Tasks are visual question-answering tasks; Position of Text predicts the spatial location of specific text; Text Recognition identifies text content given its bounding box; and VQA Grounding provides the coordinates of the answer while answering questions. To guarantee a strong connection between text and location data, we strictly maintain their alignment, ensuring that text information always comes before any associated location details.

To standardize images of different ratios, we use a scale of (0, 1000) to represent positional information. Therefore, in an image with resolutions of ($H_r \times W_r$), the text coordinates (x, y) will be normalized to $[(x/H_r * 1000)]$, and the same applies to y . The restoration process involves the inverse operation. For the input images, we resize them to 896×896 without preserving the original aspect ratio.

E. Dataset Construction

During our training process, we solely utilize open-source data and apply various task-specific prompts to different datasets. By integrating various datasets and employing different instructions for different tasks, we enhance the model’s learning ability and training efficiency. For scene text scenario, we select COCOText [34], TextOCR [35], HierText [36], TextVQA [37], and MLT [38] for training. For document images, we select IIT-CDIP [39], DocVQA [40], ChartQA [41], InfoVQA [42], DeepForm [43], Kleister Charity (KLC) [44], and WikiTable-Questions (WTQ) [45]. To accelerate the training speed, we have transformed single-image question answering into multi-turn image-based question answering, significantly improving the utilization of image features, following the successful approach

introduced in LLaVA [5]. We have a total of 409.1 k pairs of dialogue data and 2.1 M question-answering pairs in our dataset to train our model.

To further strengthen the model’s ability to handle structured text, we fine-tune one epoch on TextMonkey with structured data to enhance its structured capabilities, resulting in TextMonkey[†]. The fine-tuning data primarily consisted of 5% of the data from the previous stage, as well as a portion of structured data, including documents, tables, and charts. The structured data images are also sourced from publicly available datasets and are generated using their structure information. Therefore, we have a total of 55.7 k of data in structured data.

$$r = C_{gumbal_softmax(p)} \quad (6)$$

F. Loss

Since TextMonkey is trained to predict the next tokens like other LLMs, it only requires maximizing the likelihood of the correct next tokens (i.e., minimizing the negative log-likelihood loss $\mathcal{L}$) during training.

$$\mathcal{L} = \max \sum_{i=1}^L \log P(\tilde{s}_i | \mathbf{I}, \mathbf{Q}, \mathbf{s}_{1:i}), \quad (7)$$

where $\mathbf{I}$ is the input image, $\mathbf{Q}$ is the question sequence, $\tilde{s}$ is the output sequence, $\mathbf{s}$ is the input sequence, L is the length of the output sequence.

IV. EXPERIMENTS

A. Implementation Details

Model Configuration: In our experiments, we utilized the well-trained Vit-BigG and LLM from Qwen-VL [31], which is a pre-trained large multimodal model. We configured the height and width (H_v , W_v) of the image inputs to 448 to align with the encoder specifications of Qwen-VL. Our image resampler is equipped with 256 learnable queries, and the token resampler’s ratio (r) was set to 512 for images with a resolution of 896 and increased to 1024 for images with a resolution of 1344. To maximize training efficiency, our primary experimental focus was on using TextMonkey and evaluating outcomes at the 896 resolution setting.

TextMonkey consists of a large language model with 7.7B parameters, an image resampler module with 90 M parameters, a token resampler module with 13 M, an encoder with 1.9B parameters, and Shifted Window Attention with 45 M parameters. Overall, TextMonkey has a total of 9.7B parameters.

Training: During the training phase, we utilized the AdamW [53] optimizer, setting the learning rate to $1e-5$ for the initial stage and reducing it to $5e-6$ for the subsequent stage, while adopting a cosine learning rate schedule. The parameters β_1 and β_2 were configured to 0.9 and 0.95, respectively. A warmup period comprising 150 steps was incorporated, and we processed the data in batches of 128. To mitigate the risk of overfitting, we applied a weight decay factor of 0.1. The comprehensive training procedure spanned across 12 A800 days to complete one epoch.

Evaluation: To facilitate a more equitable comparison with other approaches, we adopted the accuracy metric, where a response produced by our model is considered correct if it encompasses the ground truth. The selection of test datasets and the formulation of evaluation criteria were carried out in accordance with the methodology described in [10]. To ensure an even fairer comparison with other methods, we also performed supplementary evaluations on certain datasets utilizing their original metrics, such as F1 score and ANLS (Average Normalized Levenshtein Similarity).

B. Results

OCRBench Results: We conduct a comparative analysis of our approach with recent large multimodal models. For our evaluation, we utilize three Scene Text-Centric VQA datasets: STVQA [54], TextVQA [37], and OCRVQA [55]; three Document-Oriented VQA datasets: DocVQA [40], InfoVQA [42], and ChartQA [41]; and three Key Information Extraction (KIE) datasets: FUNSD [56], SROIE [57], and POIE [58]. For a comprehensive assessment of performance, our evaluation includes OCRBench [10], a recent benchmark specifically developed to evaluate the Optical Character Recognition (OCR) capabilities of Large Multimodal Models. OCRBench spans a wide range of text-related visual tasks, encompassing 29 datasets, and is designed to generate an overall score.

As shown in Table II, our model demonstrates superior performance compared to existing large multimodal models, particularly in scenarios where the text is dense and small. Our method inherently enhances many current evaluation datasets, resulting in average performance improvements with numerous baseline methods by 5.2%, 6.9%, and 2.8% for Scene Text-Centric VQA, Document Oriented VQA and KIE, respectively. TextMonkey can achieve 64.3% on DocVQA and 58.2% on ChartQA. Specifically, our model achieved a score of 561 on OCRBench. The performance on both two challenging downstream tasks and OCRBench demonstrates its effectiveness in text-related tasks. Note that on the POIE dataset, our result is noticeably lower than that of InternLM-XComposer2. Our analysis indicates that the primary source of error is missing unit predictions. Our training set comprised only 409 K samples from 12 publicly available datasets, which offer limited coverage of unit-related cases. By contrast, InternLM-XComposer2 was trained on more than 30 datasets, plus a substantial volume of proprietary in-house data, which may result in better performance in a specific case.

Document Benchmarks results: To further compare and assess the capabilities of our method, we conduct tests on additional datasets utilizing the specific evaluation metric provided in their paper: F1-score for Deepform and KLC, accuracy for WTQ, relaxed accuracy measure for ChartQA, ANLS for DocVQA, and VQA score for TextVQA.

The results, shown in Table III, indicate that our model leads in performance on these datasets, outperforming other models. Across different domains, TextMonkey achieves a score of 71.5 in DocVQA, 30.6 in WTQ, 65.5 in ChartQA and 68.0 in TextVQA. It shows our model’s capability to handle documents, tables, charts, and scene text.

TABLE II
QUANTITATIVE ACCURACY (%) COMPARISON OF OUR MODEL WITH EXISTING LARGE MULTIMODAL MODELS (LMMs) ON SEVERAL BENCHMARKS. WE FINE-TUNE ON TEXTMONKEY WITH STRUCTURED DATA SHOWN IN SECTION III-E, RESULTING IN TEXTMONKEY†.

Method	STVQA	TextVQA	OCRVQA	DocVQA	InfoVQA	ChartQA	FUNSD	SROIE	POIE	OCRBench
BLIP2-OPT-6.7B [46]	20.9	23.5	9.7	3.2	11.3	3.4	0.2	0.1	0.3	235
InstructBLIP [47]	27.4	29.1	41.3	4.5	16.4	5.3	0.2	0.6	1.0	276
LLaVAR [7]	39.2	41.8	24.0	12.3	16.5	12.2	0.5	5.2	5.9	346
BLIVA [48]	32.1	33.3	50.7	5.8	23.6	8.7	0.2	0.7	2.1	291
mPLUG-Owl2 [49]	49.8	53.9	58.7	17.9	18.9	19.4	1.4	3.2	9.9	366
LLaVA1.5-7B [50]	38.1	38.7	58.1	8.5	14.7	9.3	0.2	1.7	2.5	297
TGDoc [24]	36.3	46.2	37.2	9.0	12.8	12.7	1.4	3.0	22.2	-
UniDoc [8]	35.2	46.2	36.8	7.7	14.7	10.9	1.0	2.9	5.1	-
DocPedia [13]	45.5	60.2	57.2	47.1	15.2	46.9	29.9	21.4	39.9	-
Monkey [12]	54.7	<u>64.3</u>	64.4	50.1	25.8	54.0	24.1	41.9	19.9	514
InternVL [51]	62.2	59.8	30.5	28.7	23.6	45.6	6.5	26.4	25.9	517
InternLM-XComposer2 [52]	59.6	62.2	49.6	39.7	28.6	51.6	15.3	34.2	49.3	511
TextMonkey	<u>61.8</u>	65.9	<u>71.3</u>	<u>64.3</u>	<u>28.2</u>	<u>58.2</u>	<u>32.3</u>	47.0	27.9	561
TextMonkey†	<u>61.2</u>	<u>64.3</u>	72.2	66.7	28.6	59.9	42.9	<u>46.2</u>	32.0	<u>558</u>

TABLE III
QUANTITATIVE RESULTS ON OTHER DOCUMENT BENCHMARKS. “DF” IS AN ABBREVIATION FOR DEEPFORM.

Method	Document DocVQA	DF	KLC	Table WTQ	Chart ChartQA	Scene TextVQA
Donut [3]	67.5	61.6	30.0	18.8	41.8	43.5
Pix2Struct [4]	<u>72.1</u>	-	-	-	56.0	-
UReader [11]	65.4	49.5	32.8	29.4	59.3	57.6
Qwen-VL [31]	65.1	3.1	13.9	21.6	<u>65.7</u>	63.8
Monkey [12]	66.5	40.5	<u>33.9</u>	25.3	65.1	<u>67.6</u>
TextMonkey	71.5	61.6	37.8	<u>30.6</u>	65.5	68.0
TextMonkey†	73.0	<u>59.7</u>	37.8	31.9	66.9	65.6

TABLE IV
QUANTITATIVE ACCURACY OF TEXT SPOTTING. THE “TOTAL-TEXT” AND “CTW1500” DATASETS DO NOT USE A SPECIFIC VOCABULARY FOR EVALUATION, WHILE THE “ICDAR 2015” DATASET USES A GENERAL VOCABULARY FOR EVALUATION OF OTHER MODELS. NOTE TTS ONLY USES SYNTHETIC LOCATION DATA. TEXTMONKEY IS NOT FINE-TUNED BY THE DOWNSTREAM TEXT SPOTTING DATASETS WITHOUT ANY VOCABULARY.

Method	Total-Text [59]		CTW1500 [60]		ICDAR 2015 [61]	
	Trans	Pos	Trans	Pos	Trans	Pos
TOSS [62]	61.5	65.1	51.4	54.2	47.1	52.4
TTS [63]	-	75.1	-	-	-	70.1
SPTS v2 [64]	64.7	75.5	55.4	63.6	55.6	72.6
TextMonkey	78.2	61.4	63.2	57.5	66.9	45.1

Text spotting results: To show the extensive capabilities of our model, we assessed its performance on text spotting datasets without finetuning, as detailed in Table IV. Given our model’s focus on identifying complete text passages, we segmented the predicted content into individual words for analysis. We employed two evaluation methodologies to evaluate our model’s performance. In the “Trans” mode, text is considered correct if the answer contains this word. Conversely, the “Pos” mode requires the consideration of positional information in accordance with previous methods [65]. For both metrics, due to granularity issues of the output (TextMonkey often produces an integrated paragraph while others only produce desired words), the metric can not strictly follow the evaluation setup; however, both should be quite similar, as both the error and correct situations match in calculations.

To maintain TextMonkey’s consistent performance, we refrained from fine-tuning it with downstream text spotting data, unlike other methods that were optimized for either the “Trans” or “Pos” metrics. Our results reveal that, for the “Trans” metric, TextMonkey outperformed SPTS v2 by a margin of 10.9%. Regarding the “Pos” metric, it demonstrated competent text spotting capabilities, showing its ability in understanding both text content and spatial positioning.

C. Visualization

We conduct a qualitative evaluation of TextMonkey across various scenarios, including natural scenes and document images. As shown in the left part of Fig. 5, TextMonkey accurately locates and identifies text in both scene and document images. It should be noted that in the grounding task, some words are not fully detected while their recognition results remain correct. This occurs because text recognition operates in the feature space rather than being strictly confined to the detected bounding boxes. The image tokens already encode global information, enabling the model to accurately infer the text content. A similar phenomenon has also been observed in SPTS-v2 [64] and TOSS [62]. Besides, natural images in Fig. 5(a), documents in Fig. 5(b), charts in Fig. 5(c), and tables in Fig. 5(d) exemplify TextMonkey’s adeptness at discerning and interpreting visual and textual information within a wide range of scenarios. Overall, TextMonkey’s performance across diverse scenarios demonstrates its effectiveness in perceiving and comprehending textual information in various visual contexts.

D. Ablation Study

Ablation study on zero initialization: Since CLIP is already pretrained, it is advisable to avoid drastic changes in features during the early training stages. As shown in Table V, incorporating this zero initialization method can yield 0.6% performance gain on ChartQA.

Ablation study on different components: As shown in Table VI, by introducing cross-window connections, we achieved an improvement of 0.1% on SROIE, 1.5% on DocVQA, and 2.4% on TextVQA. It can be observed that

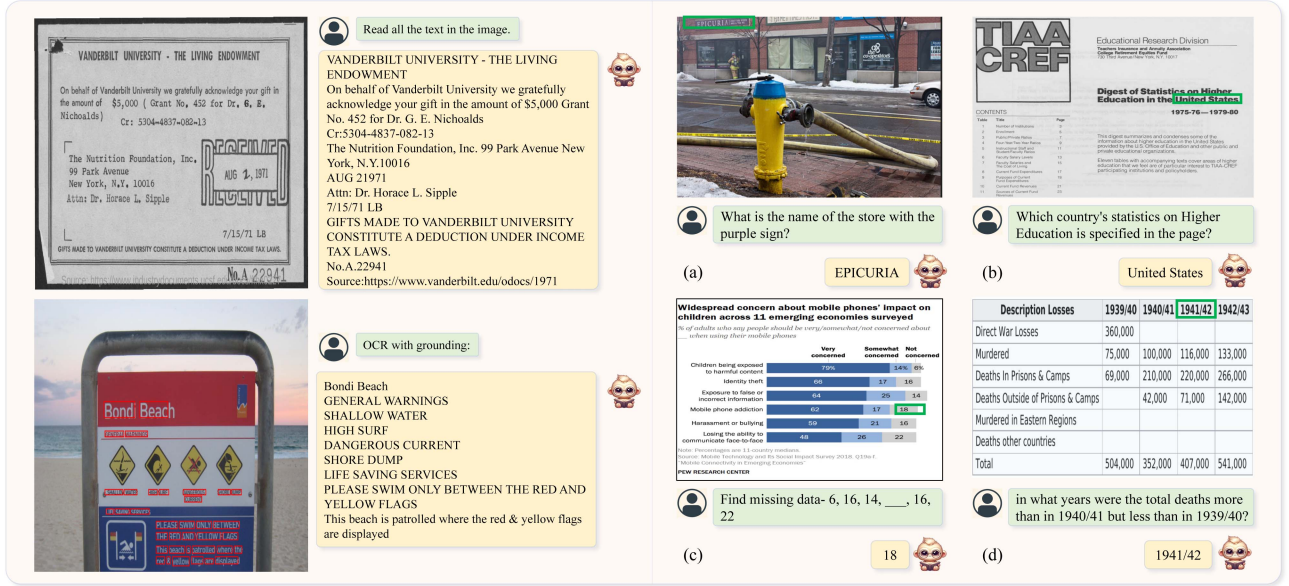


Fig. 5. Visualization results of TextMonkey. The bounding boxes generated by the model are visualized in red. The location of the ground truths are highlighted with green boxes.

TABLE V
ABLATION STUDY ON ZERO INITIALIZATION

Zero Initialization	SROIE	DocVQA	TextVQA	ChartVQA
×	46.8	64.1	65.7	57.6
✓	47.0	64.3	65.9	58.2

TABLE VI
ABLATION STUDY ON DIFFERENT COMPONENTS. "W-ATTN" MEANS SHIFTED WINDOW ATTENTION, "T-RESAMPLER" MEANS TOKEN RESAMPLER.

W-Attn	T-Resampler	SROIE	DocVQA	TextVQA
×	×	45.9	62.6	62.4
✓	×	46.0	64.1	64.8
✓	✓	47.0	64.3	65.9

cross-window connections partially compensate for the discontinuity caused by chunking and contribute to a better understanding of the images. Although Qwen2.5-VL supports native-resolution inputs without requiring image tiling that may cause text fragmentation, its training consumes 4.1 T tokens, whereas our approach uses only 409.1 k training samples. Based on the Token Resampler, our method demonstrates better performance, achieving 1.0%, 0.2%, and 1.1% performance gain on the SROIE, DocVQA, and TextVQA. This suggests that our approach effectively preserves essential information while eliminating redundant tokens, thereby simplifying the learning process for the model.

Ablation study on strategies of reducing token length: As demonstrated in Table VII, substituting important tokens with random ones (without token filter) leads to an average decline in performance by roughly 12.7%. This decline is attributed to the increased complexity of optimizing random queries, which necessitates more datasets to achieve a generalized representation compared to utilizing significant tokens. Solely focusing

TABLE VII
EFFECTIVENESS OF THE STRATEGY OF TOKEN RESAMPLER

Method	SROIE	DocVQA	TextVQA
w/o token filter	32.9	46.7	59.5
w/o resampler	44.9	63.5	62.5
w/unsorted token filter	46.8	62.1	64.2
ours	47.0	64.3	65.9

TABLE VIII
INTERACTION BETWEEN RESOLUTION AND THE NUMBER OF TOKENS REMAINED "R". "R" IN "R" MEANS DO NOT USE TOKEN RESAMPLER AND KEEP ALL THE REMAINING TOKENS.

Resolution	r	SROIE	DocVQA	TextVQA	InfoVQA
896	-	46.0	64.1	64.8	29.1
896	256	47.0	60.9	65.2	25.9
896	512	47.0	64.3	65.9	28.2
1344	-	42.9	54.9	62.5	28.9
1344	512	44.9	59.7	64.2	28.0
1344	1024	46.0	64.5	65.1	31.4

on pivotal features (without resampler) and directly eliminating features incurs a loss of some information, showing a decrease in performance, such as a 2.1% drop in SROIE. Additionally, neglecting the order of tokens (with unsorted token filter) does not markedly impair performance, owing to the language model's inherent ability to organize unordered tokens. Nevertheless, the lack of token order can still lead to decrease, especially evident in the result of DocVQA, with a 2.2% decrease in performance.

Interaction between input resolution and the number of tokens remained: As shown in Table VIII, directly increasing the resolution without compressing tokens can actually lead to consistent worse performance, especially with a decrease of 9.2% performance in DocVQA. We speculate that the increase in resolution results in a significant increase in redundant tokens, making it more difficult to find crucial information in our

TABLE IX
ABLATION STUDY ON EACH AUGMENTED VQA DATASET

Excluded Datasets	SROIE	DocVQA	TextVQA	ChartVQA	Average
None	47.0	64.3	65.9	58.2	58.9
Read All Text	45.5	63.0	63.9	57.3	57.4
Text Spotting	48.3	64.7	63.8	58.1	58.7
Original Tasks	45.4	63.3	63.0	55.3	56.8
Text Recognition	47.3	65.6	64.0	58.2	58.8
VQA Grounding	48.3	64.6	64.5	58.0	58.9

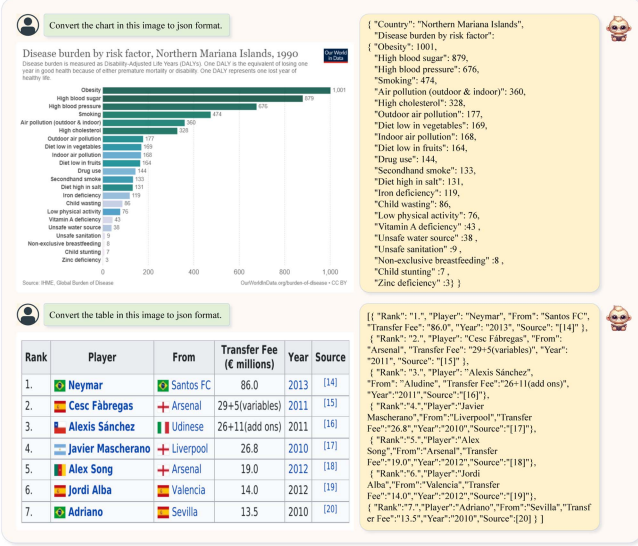


Fig. 6. Examples of structuralization of chart and table using TextMonkey†.

setting. Therefore, compressing tokens reasonably can lead to higher performance. Considering the sparsity of information in large-sized images, it is also necessary to consider selecting an appropriate value of “r” for different input resolutions. Besides, increasing the input resolution brings benefits to the dataset, which contains many large-sized images, with 0.2% performance gain for DocVQA and 3.2% performance gain for InfoVQA. However, for datasets like TextVQA and SROIE, which contain much smaller images, increasing the input resolution directly does not yield any gains.

Ablation study on each augmented VQA dataset: We further conducted ablation studies on each augmented VQA dataset, as shown in Table IX. The results indicate that most augmented VQA datasets have a positive effect on the model’s overall performance. Additionally, incorporating the VQA Grounding task enables the model to provide answers along with their corresponding locations, which facilitates a better analysis of the reasons behind the model’s errors.

E. Structuralization

The structuralization of charts and tables holds substantial practical value. Structured charts and tables present data in a clear format, and by extracting structural information from images, computers can accurately parse and extract the data. This makes data analysis, statistics, and modeling more efficient and precise. It also helps reduce the complexity of information and improves its comprehensibility. As depicted in Fig. 6, our

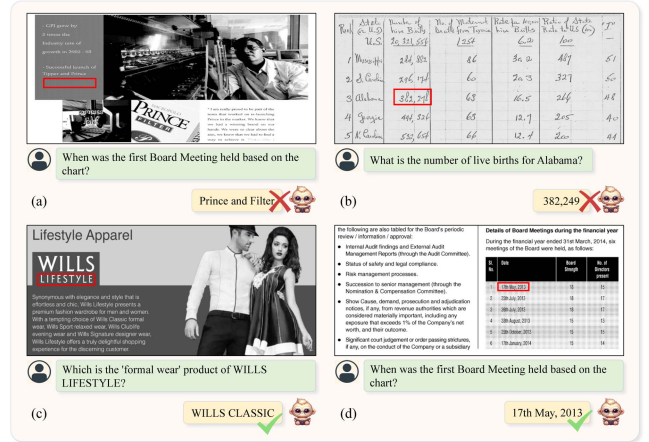


Fig. 7. Examples of grounding the position of the answer.

model is capable of structuring charts and tables into JSON format, demonstrating its potential for downstream applications. According to Table III, TextMonkey† exhibits a performance improvement of 1.3% and 1.4% on tables and charts, respectively. This underscores that high-quality data not only enables the model’s structuralization capabilities but also amplifies the effectiveness of the related benchmarks. However, it is worth noting that this type of data will primarily benefit the data within its own domain, thus leading to a performance decrease for cross-domain TextVQA.

V. DISCUSSION

A. Interpretability

By examining the grounding information, we can identify the reasons behind the model’s errors, enhancing a better understanding of the model’s behavior. As shown in Fig. 7(a), we ground to the white region, indicating that the model might be engaging in hallucination. We correctly identify the location but recognize it wrongly in Fig. 7(b). Fig. 7(c) highlights a scenario where the model grounds to incorrect text but still provides the correct answer. This could imply that there is some level of randomness or uncertainty in the model’s responses at this point. In Fig. 7(d), the alignment between the position and text indicates that the model is more confident in its predictions. Therefore, based on these analyses, we can gain a better understanding of the model’s behavior and have a better awareness of the model’s hallucination, thus reducing the model’s hallucination.

B. Chain-of-Thought

We also conduct experiments on several datasets and observe inconsistent improvements if we require a model to provide the answer’s position, as shown in Table XI. In datasets where the majority of answers are based on information within the images, such as DocVQA and SROIE, there is a noticeable benefit in requiring the model to provide the answer’s position. However, for datasets that involve reasoning tasks, such as ChartQA and InfoVQA, where questions require comparisons or quantitative analysis (e.g., “How much more is A than B?”), demanding

TABLE X
THE EVALUATION RESULTS OF TEXTMONKEY ON OMNIDOCBENCH

Model	Training Data	Overall ^{Edit} ↓		Text ^{Edit} ↓		Formula ^{Edit} ↓		Table ^{TEDS} ↑		Table ^{Edit} ↓		Read Order ^{Edit} ↓	
		EN	ZH	EN	ZH	EN	ZH	EN	ZH	EN	ZH	EN	ZH
Qwen2.5-VL-7B [25]	-	0.312	0.406	0.157	0.228	0.351	0.574	76.4	72.2	0.588	0.619	0.149	0.203
InternVL3-8B [67]	-	0.314	0.383	0.134	0.218	0.417	0.563	66.1	73.1	0.586	0.564	0.118	0.186
TextMonkey	0.544 M	0.201	0.318	0.098	0.355	0.367	0.478	70.9	67.2	0.258	0.271	0.080	0.169
	0.640 M	0.186	0.307	0.077	0.348	0.376	0.457	71.2	66.8	0.223	0.269	0.070	0.156
TextMonkey	1.088 M	0.179	0.287	0.080	0.286	0.362	0.465	74.9	70.4	0.205	0.244	0.068	0.153

TABLE XI
EFFECT OF INCORPORATING THE POSITION OF ANSWER

Method	DocVQA	SROIE	ChartQA	InfoVQA
w position	64.5	47.2	57.8	27.7
w/o position	64.3	47.0	58.2	28.2

positional answers can actually result in a detrimental effect. Upon further examination of the wrong answer, we consider that the requirement of grounding might have partially affected certain reasoning needs. Hence, it is essential to consider the nature of the dataset and the type of questions being asked when deciding whether to impose the requirement of positional answers.

Additionally, we believe that automating the process of constructing a thinking chain [67] in subsequent steps could be a promising direction for future research. By developing mechanisms to generate a coherent chain of reasoning automatically, we can potentially enhance the overall performance and reasoning capabilities of our models.

C. Document Parsing

To further investigate the potential of TextMonkey on document parsing tasks, we conduct additional experiments on OmniDocBench [68]. On this benchmark, state-of-the-art methods are typically trained on large-scale document parsing datasets; for instance, MinerU2.5 is trained on 16.9 M samples. Due to limited time and computational resources, we train TextMonkey using only a subset of the data [69], ranging from 0.544 M to 1.088 M samples. As shown in Table X, when trained with 0.544 M samples, TextMonkey already outperforms Qwen2.5-VL-7B and InternVL3-8B by an average margin of 9.95% and 8.9%, respectively. With further increases in training data, TextMonkey achieves an additional 2.65% performance gain. These results demonstrate that, with sufficient training data, TextMonkey has the potential to reach competitive performance for document parsing.

VI. CONCLUSION

This paper introduces an LLM-based document understanding model TextMonkey. We adopt Shifted Window Attention with zero initialization to better capture spatial relationships while scaling input resolution through a sliding-window mechanism. As increasing resolution inevitably expands the number of visual tokens, we further analyze token redundancy and introduce a Token Resampler module to effectively reduce

unnecessary tokens without compromising semantic fidelity. Moreover, by engaging in multiple text-oriented tasks simultaneously, TextMonkey enhances its understanding of spatial structures within documents, thereby improving interpretability. Comparative experiments across multiple large multimodal models demonstrate that TextMonkey achieves consistently strong performance on several benchmarks. Importantly, we also observe that simply increasing input resolution does not always yield better results, especially for smaller document images, highlighting the necessity of designing efficient resolution-scaling mechanisms. Moreover, this work reveals that in document understanding, visual representations themselves can serve as an alternative and effective form of text storage and reasoning, suggesting a deeper interplay between visual and linguistic modalities in document intelligence.

REFERENCES

- [1] Z. Tang et al., "Unifying vision, text, and layout for universal document processing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 19254–19264.
- [2] Y. Huang, T. Lv, L. Cui, Y. Lu, and F. Wei, "LayoutLMv3: Pre-training for document AI with unified text and image masking," in *Proc. MM '22: 30th ACM Int. Conf. Multimedia*, Lisboa, Portugal, 2022, pp. 4083–4091.
- [3] G. Kim et al., "OCR-Free document understanding transformer," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 498–517.
- [4] K. Lee et al., "Pix2Struct: Screenshot parsing as pretraining for visual language understanding," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 18893–18912.
- [5] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *Adv. Neural Inf. Process. Syst.*, vol. 36, pp. 34892–34916, 2023.
- [6] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," in *Proc. Int. Conf. Learn. Representations*, 2024, pp. 18378–18394.
- [7] Y. Zhang et al., "LLaVAR: Enhanced visual instruction tuning for text-rich image understanding," 2023, *arXiv:2306.17107*.
- [8] H. Feng et al., "UniDoc: A universal large multimodal model for simultaneous text detection, recognition, spotting and understanding," 2023, *arXiv:2308.11592*.
- [9] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [10] Y. Liu et al., "OCRBench: On the hidden mystery of OCR in large multimodal models," *Sci. China Inf. Sci.*, vol. 67, no. 12, 2024, Art. no. 220102.
- [11] J. Ye et al., "UReader: Universal OCR-Free visually-situated language understanding with multimodal large language model," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2023, pp. 2841–2858.
- [12] Z. Li et al., "Monkey: Image resolution and text label are important things for large multi-modal models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 26763–26773.
- [13] H. Feng et al., "DocPedia: Unleashing the power of large multimodal model in the frequency domain for versatile document understanding," *Sci. China Inf. Sci.*, vol. 67, no. 12, 2024, Art. no. 220106.
- [14] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 10012–10022.

- [15] Y. Li et al., "StrucTexT: Structured text understanding with multi-modal transformers," in *Proc. 29th ACM Int. Conf. Multimedia*, 2021, pp. 1912–1920.
- [16] Q. Peng et al., "ERNIE-Layout: Layout knowledge enhanced pre-training for visually-rich document understanding," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 3744–3756.
- [17] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, "LayoutLM: Pre-training of text and layout for document image understanding," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2020, pp. 1192–1200.
- [18] Y. Xu et al., "LayoutLMv2: Multi-modal pre-training for visually-rich document understanding," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 2579–2591.
- [19] H. Bai et al., "Wukong-reader: Multi-modal pre-training for fine-grained visual document understanding," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguistics*, 2023, pp. 13386–13401.
- [20] S. Appalaraju, P. Tang, Q. Dong, N. Sankaran, Y. Zhou, and R. Manmatha, "DocFormerv2: Local features for document understanding," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 2, pp. 709–718.
- [21] B. Davis, B. Morse, B. Price, C. Tensmeyer, C. Wington, and V. Morariu, "End-to-end document recognition and understanding with dessurt," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 280–296.
- [22] Y. Yu et al., "StrucTexTv2: Masked visual-textual prediction for document image pre-training," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, pp. 1–12.
- [23] H. Wei et al., "Vary: Scaling up the vision vocabulary for large vision-language models," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 408–424.
- [24] Y. Wang, W. Zhou, H. Feng, K. Zhou, and H. Li, "Towards improving document understanding: An exploration on text-grounding via mllms," 2023, *arXiv:2311.13194*.
- [25] S. Bai et al., "Qwen2.5-VL technical report," 2025, *arXiv:2502.13923*.
- [26] S. Bai et al., "Qwen3-VL technical report," 2025, *arXiv:2511.21631*.
- [27] Y. Li, G. Yang, H. Liu, B. Wang, and C. Zhang, "dots.ocr: Multilingual document layout parsing in a single vision-language model," 2025, *arXiv:2512.02498*.
- [28] J. Niu et al., "MinerU2.5: A decoupled vision-language model for efficient high-resolution document parsing," 2025, *arXiv:2509.22186*.
- [29] H. Wei, Y. Sun, and Y. Li, "DeepSeek-OCR: Contexts optical compression," 2025, *arXiv:2510.18234*.
- [30] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. 10th Int. Conf. Learn. Representations*, 2022, pp. 1–13.
- [31] J. Bai et al., "Qwen-vl: A frontier large vision-language model with versatile abilities," 2023, *arXiv:2308.12966*.
- [32] D. Bolya, C.-Y. Fu, X. Dai, P. Zhang, C. Feichtenhofer, and J. Hoffman, "Token merging: Your ViT but faster," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, pp. 1–12.
- [33] J.-B. Alayrac et al., "Flamingo: A visual language model for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 23716–23736.
- [34] A. Veit, T. Matera, L. Neumann, J. Matas, and S. Belongie, "COCO-Text: Dataset and benchmark for text detection and recognition in natural images," Jan. 2016, *arXiv:1601.07140*.
- [35] A. Singh, G. Pang, M. Toh, J. Huang, W. Galuba, and T. Hassner, "TextOCR: Towards large-scale end-to-end reasoning for arbitrary-shaped scene text," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 8802–8812.
- [36] S. Long, S. Qin, D. Panteleev, A. Bissacco, Y. Fujii, and M. Raptis, "Towards end-to-end unified scene text detection and layout analysis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 1049–1059.
- [37] A. Singh et al., "Towards VQA models that can read," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 8317–8326.
- [38] N. Nayef et al., "ICDAR2019 robust reading challenge on multi-lingual scene text detection and recognition—RRC-MLT-2019," in *Proc. 2019 Int. Conf. Document Anal. Recognit.*, 2019, pp. 1582–1587.
- [39] D. Lewis, G. Agam, S. Argamon, O. Frieder, D. Grossman, and J. Heard, "Building a test collection for complex document information processing," in *Proc. 29th Annu. Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2006, pp. 665–666.
- [40] M. Mathew, D. Karatzas, and C. Jawahar, "DocVQA: A dataset for VQA on document images," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2021, pp. 2200–2209.
- [41] A. Masry, X. L. Do, J. Q. Tan, S. Joty, and E. Hoque, "ChartQA: A benchmark for question answering about charts with visual and logical reasoning," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 2263–2279.
- [42] M. Mathew, V. Bagal, R. Tito, D. Karatzas, E. Valveny, and C. V. Jawahar, "InfographicVQA," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jan. 2022, pp. 1697–1706. [Online]. Available: https://openaccess.thecvf.com/content/WACV2022/html/Mathew_InfographicVQA_WACV_2022_paper.html?utm_source=chatgpt.com
- [43] S. Svetlichnaya, "DeepForm: Understand structured documents at scale," *Weights & Biases report*, 2020. [Online]. Available: https://wandb.ai/stacey/deepform_v1/reports/DeepForm-Understand-Structured-Documents-at-Scale--VmlldzoyODQ3Njg
- [44] T. Stanisławek et al., "Kleister: Key information extraction datasets involving long documents with complex layouts," in *Proc. Int. Conf. Document Anal. Recognit.*, 2021, pp. 564–579.
- [45] P. Pasupat and P. Liang, "Compositional semantic parsing on semi-structured tables," in *Proc. Annu. Meeting Assoc. Comput. Linguistics*, 2015, pp. 1470–1480.
- [46] J. Li, D. Li, S. Savarese, and S. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.
- [47] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," in *Proc. 37th Conf. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 49250–49267.
- [48] W. Hu, Y. Xu, Y. Li, W. Li, Z. Chen, and Z. Tu, "BLIVA: A simple multimodal LLM for better handling of text-rich visual questions," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 3, pp. 2256–2264.
- [49] Q. Ye et al., "mPLUG-Owl2: Revolutionizing multi-modal large language model with modality collaboration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 13040–13051.
- [50] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 26296–26306.
- [51] Z. Chen et al., "InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24185–24198.
- [52] X. Dong et al., "InternLM-XComposer2: Mastering free-form text-image composition and comprehension in vision-language large model," 2024, *arXiv:2401.16420*.
- [53] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Representations*, 2019, pp. 1–10.
- [54] A. F. Biten et al., "Scene text visual question answering," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4291–4301.
- [55] A. Mishra, S. Shekhar, A. K. Singh, and A. Chakraborty, "OCR-VQA: Visual question answering by reading text in images," in *Proc. 2019 Int. Conf. Document Anal. Recognit.*, 2019, pp. 947–952.
- [56] G. Jaume, H. K. Ekenel, and J.-P. Thiran, "FUNSD: A dataset for form understanding in noisy scanned documents," in *Proc. Int. Conf. Document Anal. Recognit. Workshops*, 2019, pp. 1–6.
- [57] Z. Huang et al., "ICDAR 2019 competition on scanned receipt OCR and information extraction," in *Proc. Int. Conf. Document Anal. Recognit.*, 2019, pp. 1516–1520.
- [58] J. Kuang et al., "Visual information extraction in the wild: Practical dataset and end-to-end solution," in *Proc. Int. Conf. Document Anal. Recognit.*, 2023, pp. 36–53.
- [59] C. K. Ch'ng and C. S. Chan, "Total-Text: A comprehensive dataset for scene text detection and recognition," in *Proc. 14th IAPR Int. Conf. Document Anal. Recognit.*, 2017, pp. 935–942.
- [60] Y. Liu, L. Jin, S. Zhang, C. Luo, and S. Zhang, "Curved scene text detection via transverse and longitudinal sequence connection," *Pattern Recognit.*, vol. 90, pp. 337–345, 2019.
- [61] D. Karatzas et al., "ICDAR 2015 competition on robust reading," in *Proc. 13th Int. Conf. Document Anal. Recognit.*, 2015, pp. 1156–1160.
- [62] J. Tang, S. Qiao, B. Cui, Y. Ma, S. Zhang, and D. Kanoulas, "You can even annotate text with voice: Transcription-only-supervised text spotting," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 4154–4163.
- [63] Y. Kittenplon, I. Lavi, S. Fogel, Y. Bar, R. Manmatha, and P. Perona, "Towards weakly-supervised text spotting using a multi-task transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 4604–4613.
- [64] Y. Liu et al., "SPTS v2: Single-point scene text spotting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15665–15679, Dec. 2023.
- [65] Y. Liu et al., "ABCNet v2: Adaptive Bezier-curve network for real-time end-to-end text spotting," vol. 44, no. 11, pp. 8048–8064, 2022.
- [66] J. Zhu et al., "InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models," 2025, *arXiv:2504.10479*.
- [67] J. Wei et al., "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, vol. 35, pp. 24824–24837.

- [68] L. Ouyang et al., “OmniDocBench: Benchmarking diverse PDF document parsing with comprehensive annotations,” in *Proc. Comput. Vis. Pattern Recognit. Conf.*, 2025, pp. 24838–24 848.
- [69] Z. Li et al., “MonkeyOCR: Document parsing with a structure-recognition-relation triplet paradigm,” 2025, *arXiv:2506.05218*.



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